

Injective.
one-one function:

$$f: \textcircled{A} \xrightarrow{\text{Domain}} \textcircled{B} \xleftarrow{\text{Codomain}}$$

for all $x_1, x_2 \in A$, If $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.

equivalently, If $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.

Ex: $f: \mathbb{Z} \rightarrow \mathbb{Z}$
 $f(n) = 2n + 1$ } injective?

$n_1, n_2 \in \mathbb{Z}$ (Domain).

$$f(n_1) = f(n_2)$$

$$\Rightarrow 2n_1 + 1 = 2n_2 + 1$$

$$\Rightarrow 2n_1 = 2n_2$$

$$\Rightarrow n_1 = n_2$$

Ex: $f: \textcircled{\mathbb{Z}} \rightarrow \mathbb{Z}$

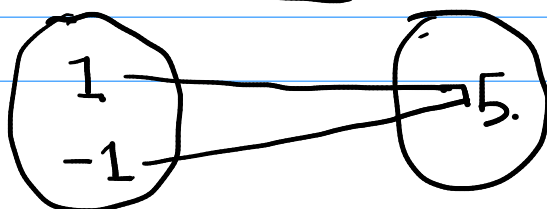
$$f(n) = 3n^2 + 2$$
 } injective?

$x_1 = 1$
 $x_2 = 3$

$$f(n) = 3n^2 + 2 = f(-n)$$

$$x_1, x_2$$

$$\underline{f(1) = 5 = f(-1)}$$



$$f: \mathbb{R} \rightarrow \mathbb{R}.$$

$$f(x) = x^2.$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$x \rightarrow |x|.$$

$$1 \rightarrow 1$$

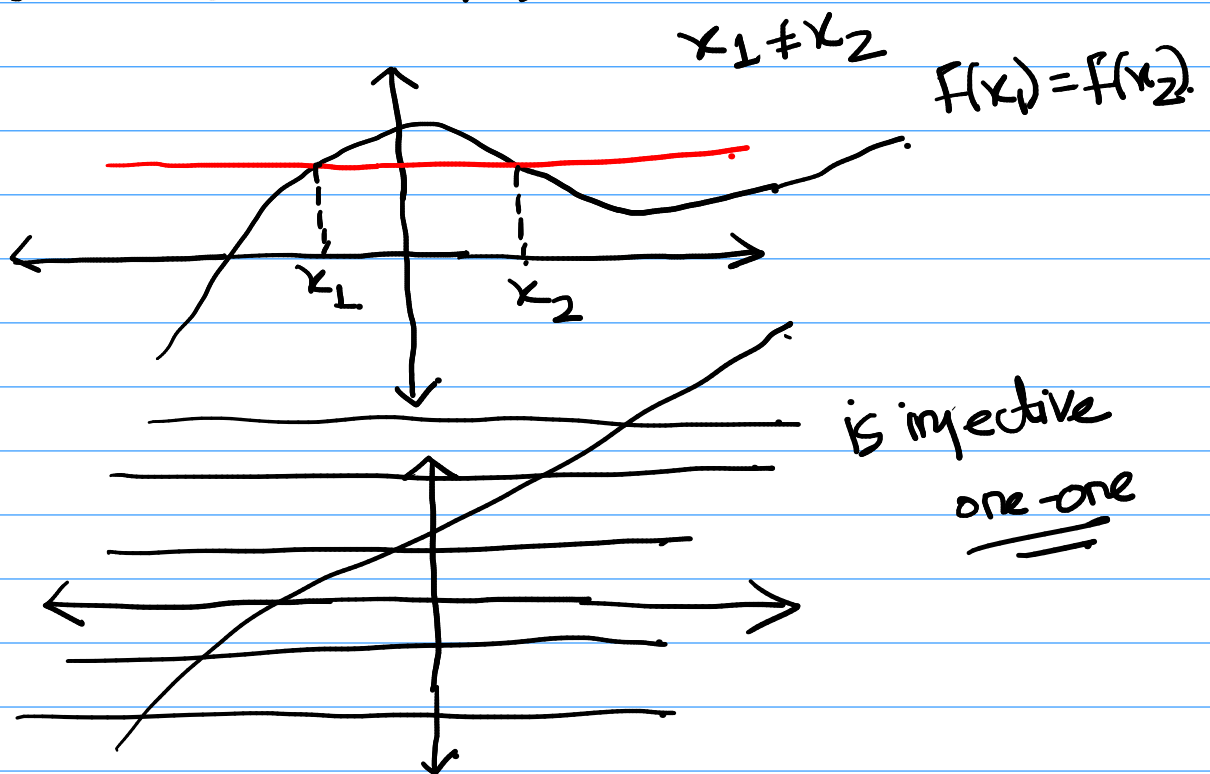
$$-1 \rightarrow 1.$$

$$\bullet \quad f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$$

$$x \rightarrow 1/x. \text{ - injective}$$

$$x_1 \neq x_2 \Rightarrow 1/x_1 \neq 1/x_2 \Rightarrow f(x_1) \neq f(x_2).$$

• Horizontal line test:



$$f: \mathbb{R} \rightarrow \mathbb{R}.$$

$$\bullet \quad f(x) = x^3 - x.$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = \sin x.$$

$$\bullet \quad ax^2 + bx + c.$$

$$\boxed{\log(x) \neq \underline{ax^2 + bx + c}}$$

$$\ln(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} (x-1)^n}{n}$$

↑

$$f: (A) \rightarrow (B)$$

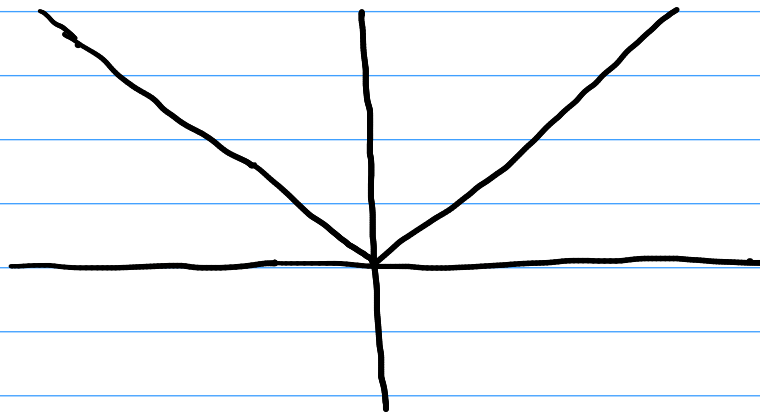
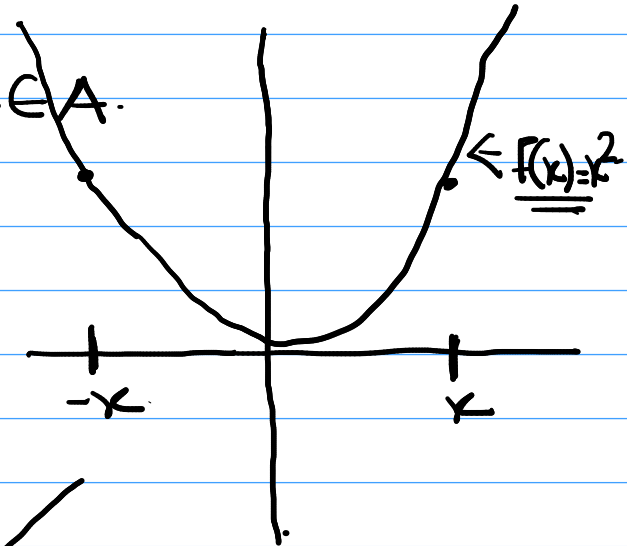
Even-odd functions

$$f: A \rightarrow B$$

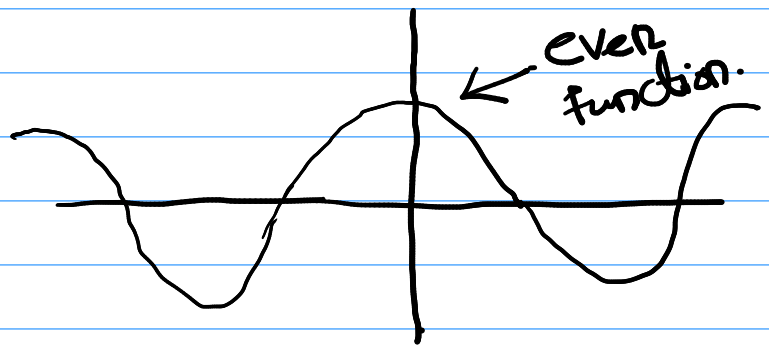
Even: $f(-x) = f(x) \quad \forall x \in A.$

$$f(-x) = (-x)^2 = x^2 = f(x)$$

$$f(x) = |x|.$$

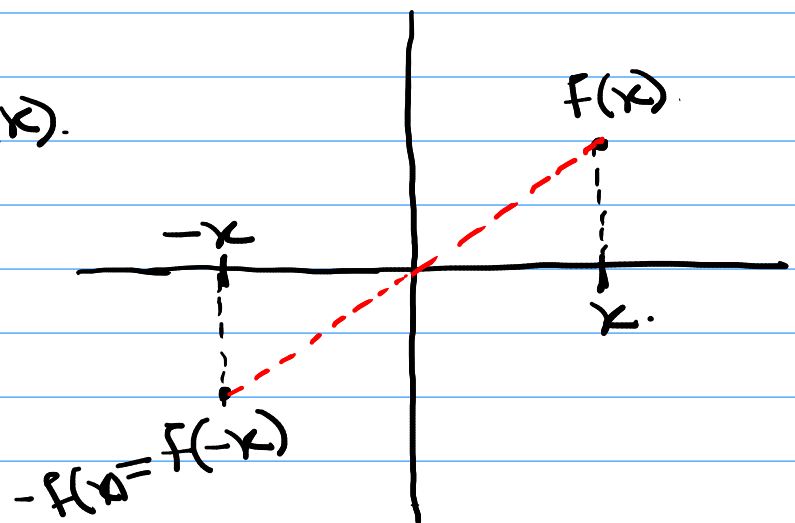


• $f(x) = \cos x.$



odd-function

$$f(-x) = -f(x).$$



$f(x) = \sin x$ — odd function.

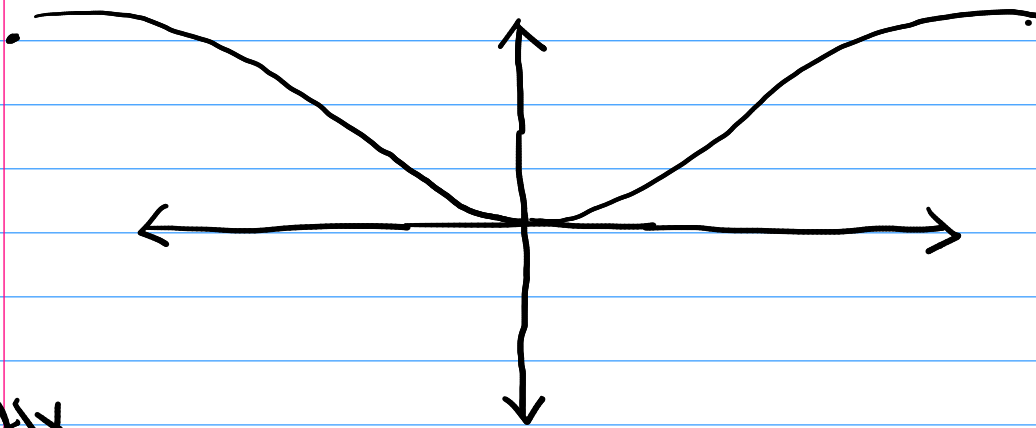
$$f(x) = x^3$$

$$f(-x) = (-x)^3 = -x^3 = -f(x).$$

Ex: $f(x) = 3x^2 - 4 \leftarrow$ even function

$$f(x) = x^3 - x^2 \leftarrow \text{neither even. nor } \underline{\underline{\text{odd}}}$$

$$f(-x) = (-x)^3 - (-x)^2 = -x^3 - x^2 \stackrel{?}{=} f(x) \\ \stackrel{?}{=} -f(x)$$



Strictly

Increasing function // decreasing function.

If $\overset{\downarrow \text{Dom}(f)}{x_1} > x_2 \Rightarrow f(x_1) > f(x_2).$

If $x_1 > x_2 \Rightarrow f(x_1) < f(x_2).$

Ex: $f(x) = 1 -$ Increasing function
 $\rightarrow$ decreasing function.

$f(x_1) > f(x_2)$

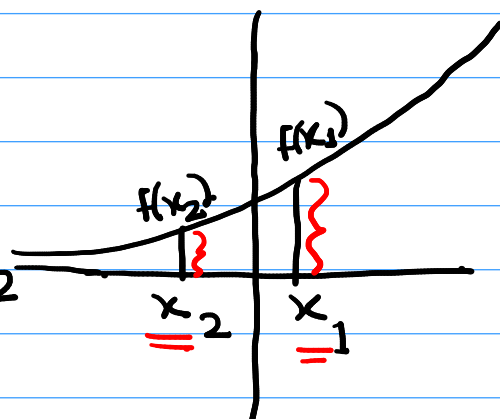
$$f(x) = x + 1$$

$$\text{If } x_1 > x_2 \Rightarrow x_1 + 1 > x_2 + 1 \\ \Rightarrow f(x_1) > f(x_2)$$

$$f(x) = a^x \quad \text{If } \underline{\underline{a > 1}}$$

$$f(x) = 2^x$$

$$x_1 > x_2 \\ \Rightarrow 2^{x_1} > 2^{x_2}$$



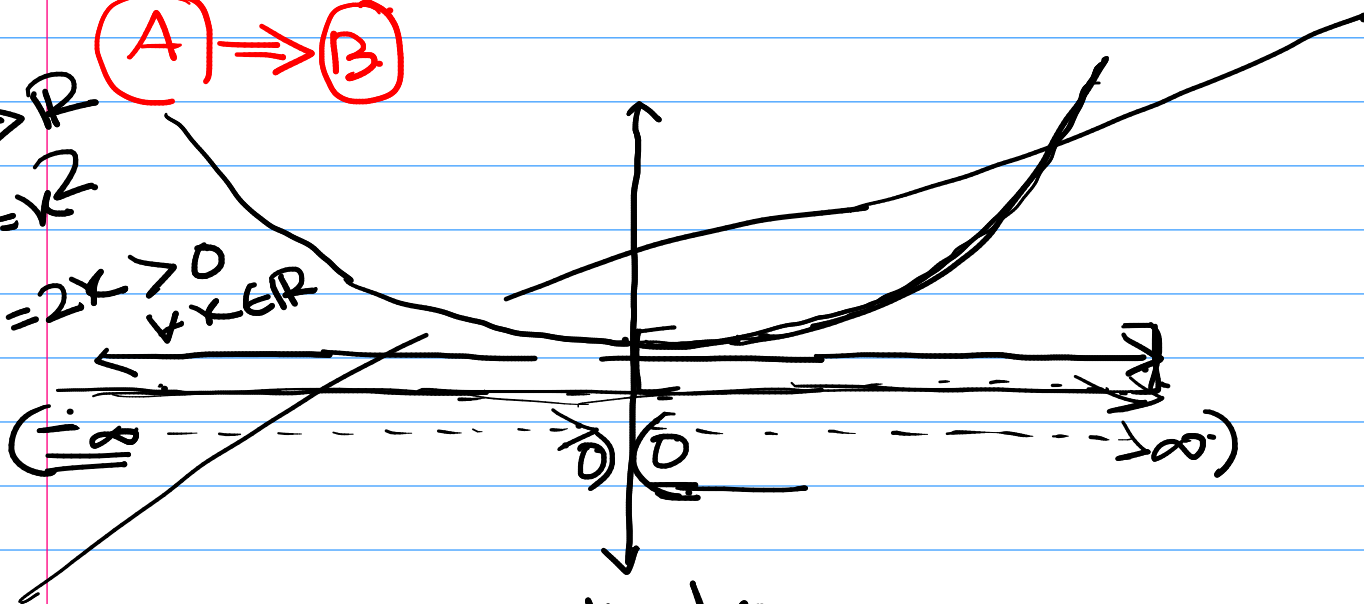
One-one functions.

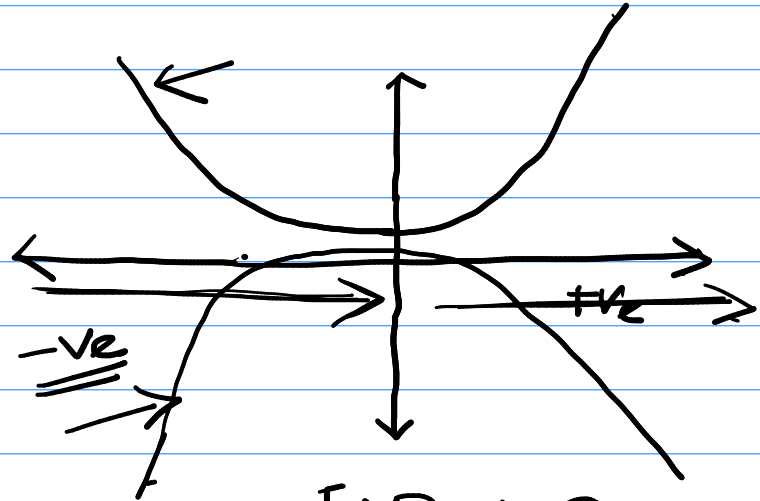
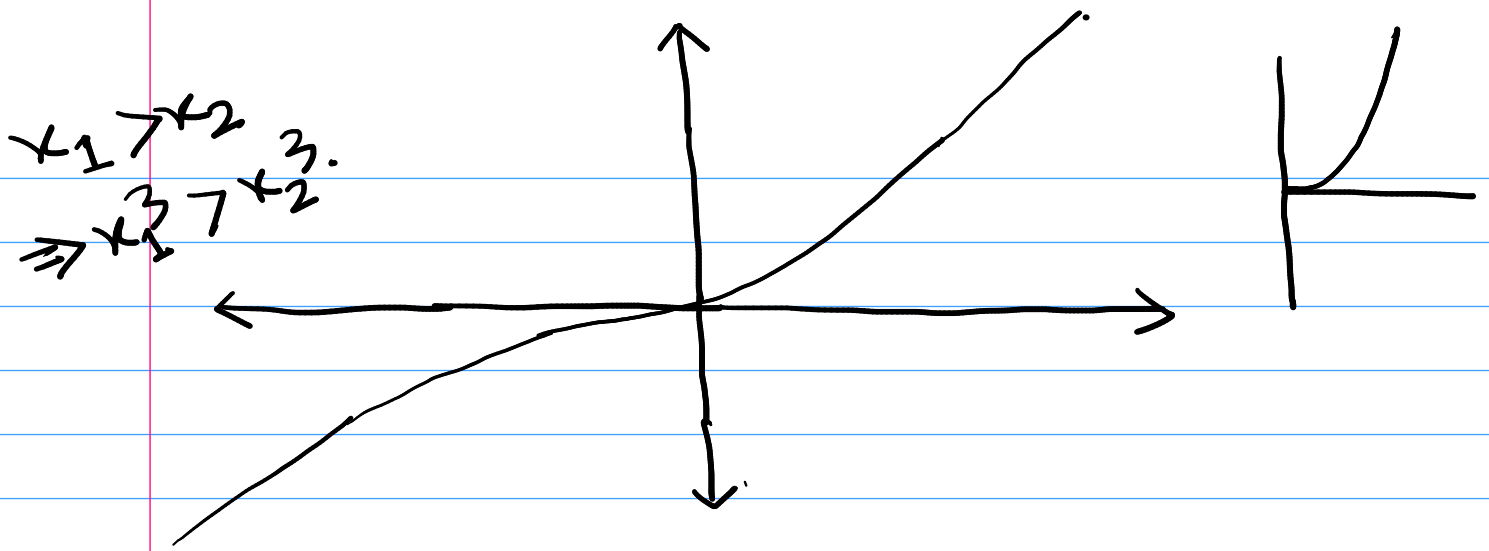
(A) $\Rightarrow$ (B)

$$\mathbb{R}^+ \rightarrow \mathbb{R}$$

$$f(x) = x^2$$

$$f'(x) = 2x > 0 \quad \forall x \in \mathbb{R}^+$$





$$f: \mathbb{R} \rightarrow \mathbb{R}.$$

$$f(x) = \lfloor x \rfloor.$$

$$f: A \rightarrow B$$

If $f'(x) > 0 \quad \forall x \in A \Rightarrow f$ is strictly increasing

If $f'(x) < 0 \quad \forall x \in A \Rightarrow f$ is strictly decreasing on A.

$$\begin{cases} f(x) = e^x & f'(x) = e^x > 0 \Rightarrow \\ f(x) = e^{-x} & f'(x) = -e^{-x} < 0 \Rightarrow \end{cases}$$

$$f: \underbrace{\{1, 2\}}_{(-3, 3)} \rightarrow \{2, 3\}$$

$$\underline{(1, 2)}$$

$$\mathbb{R}$$

$$\underline{(-3, 3)}$$

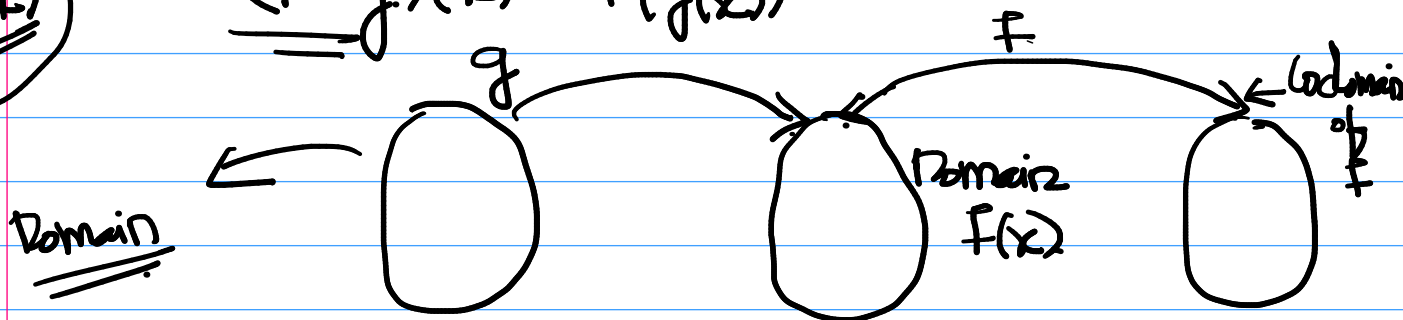
- Composition of functions

$$f: A \rightarrow B$$

$$g: C \rightarrow D$$

$$(f \circ g)(x) = f(g(x))$$

$$\underline{\underline{f(g(x))}}$$



- $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$

$$g(x): \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto \frac{1}{x-1}$$

$$x \mapsto x$$

$$(f \circ g)(1) = f(g(1)) = \underline{\underline{f(1)}}$$

- $f: \underline{[-5, 7]}$ Domain of f

$$g(x) = |2x+5|$$

$$\begin{matrix} \text{Interval} \\ [a, b] \\ \underline{\underline{(f \circ g)(x)}} \end{matrix}$$

$$\begin{matrix} \underline{\underline{x \in \mathbb{R}}} \\ \uparrow \\ (f \circ g)(x) \\ \uparrow \\ f(g(x)) \\ \uparrow \\ g(x) \in \text{Dom}(f) \end{matrix}$$

$$\begin{matrix} \text{When } g(x) \in \text{Domain of } f \\ \Rightarrow \underline{\underline{f(g(x))}} \Rightarrow \underline{\underline{|2x+5| \in [-5, 7]}} \end{matrix}$$

$$-5 \leq |2x+5| \leq 7$$

one-case

$$-5 \leq 2x+5 \leq 7$$

$$\Rightarrow -10 \leq 2x \leq 2$$

$$\Rightarrow -5 \leq x \leq 1$$

$$x \in [-5, 1]$$

$$[-5, 1]$$

$$-5$$

↙

$$f: \{1\} \rightarrow \mathbb{R}$$

$$(f \circ g)(x) \quad \forall x \in \mathbb{R}$$

$$g: \mathbb{R} \rightarrow \{1\}$$

$$x \mapsto 1$$

Inverse of a function: Range(f)

$f: A \rightarrow B$ which is one-one & on-to.

$$f^{-1}: B \rightarrow A$$

$$(f \circ f^{-1})(x) = x \quad \& \quad (f^{-1} \circ f)(x) = x$$

$$\text{Dom}(\bar{f}^{-1}) = \text{Range of } f \quad // \quad \text{Dom}(f) = \underline{\underline{\text{Range of } (\bar{f}^{-1})}}$$

- $\bar{f}^{-1}(x) \neq \frac{1}{f(x)}$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$\underline{\underline{f(x) = x}} \leftarrow \text{one-one}$$

$$\hookrightarrow \bar{f}^{-1}(x) = x \neq \frac{1}{x}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

- $f(x) = 3x - 5$
 - one-one function
 - $\text{Range}(f) = \mathbb{R}$

$$g = \bar{f}^{-1}: \mathbb{R} \rightarrow \mathbb{R}$$

$$\underline{\underline{(f \circ g)(x) = x}}$$

$$\Rightarrow f(\underbrace{g(x)}_y) = x$$

$$\Rightarrow f(y) = x$$

$$\Rightarrow 3y - 5 = x$$

$$\Rightarrow y = \frac{x+5}{3} \Rightarrow \underline{\underline{g(x) = \frac{x+5}{3}}}$$

- $\circ \quad \underline{\underline{(g \circ f)(x) = x}}$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

- $f(x) = x^4 + 7$

$\downarrow$
Not one-one

$$f: \mathbb{R} \setminus \{-2/3\} \rightarrow \mathbb{R}.$$

$$f(x) = \frac{3x+1}{3x+2}.$$

Range of $f(x)$ $\equiv D$

$$f(x) = f(y) \Rightarrow x = y.$$

$$\Rightarrow \frac{3x+1}{3x+2} = \frac{3y+1}{3y+2} \Rightarrow (3x+1)(3y+2) = (3y+1)(3x+2)$$

$$f^{-1}: D \rightarrow \mathbb{R} \setminus \{-2/3\}$$

$$(f \circ g)(x) = x$$

$$\Rightarrow f(g(x)) = x \Rightarrow f(y) = x$$

$$\Rightarrow \frac{y+1}{3y+2} = x$$

$$\Rightarrow x(3y+2) = y+1$$

$$\Rightarrow 3yx + 2x = y + 1 \Rightarrow 3yx - y = 1 - 2x$$

$$\Rightarrow y(3x-1) = 1-2x$$

$$\Rightarrow y = \frac{1-2x}{3x-1}$$

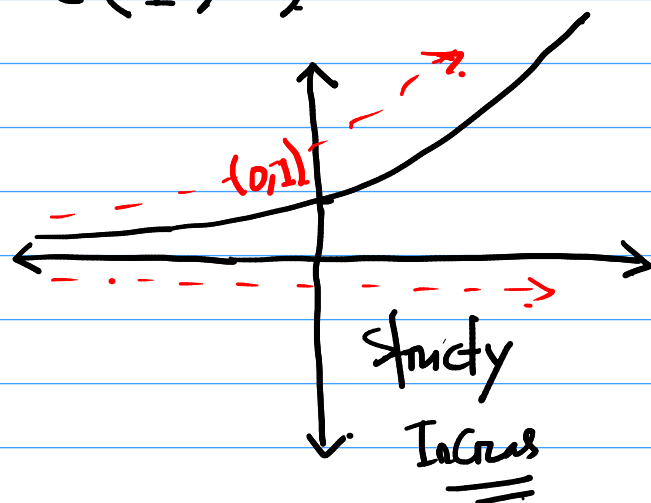
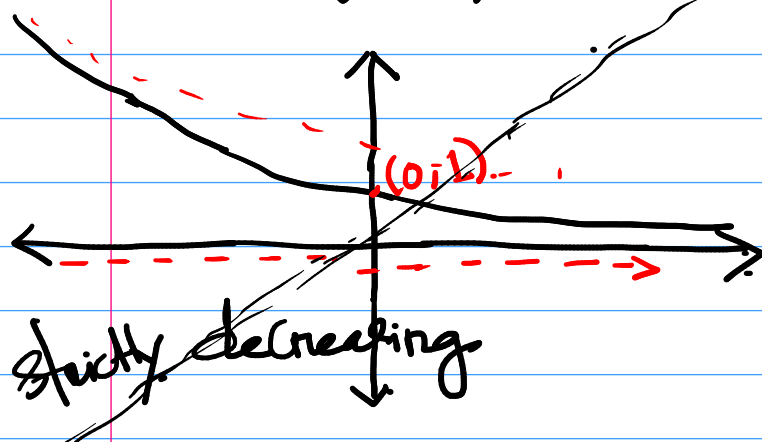
$$g(x) = \frac{1-2x}{3x-1}$$

$$\underline{\underline{\mathbb{R} \setminus \{1/3\}}} \leftarrow \text{Range of } f$$

• $f(x) = \underline{a^x}$ where $a > 0 \neq a \neq 1$

$a \in (0, 1)$

$a \in (1, \infty)$



- Domain: $\mathbb{R}$
- Range: $(0, \infty)$ $\rightarrow$ +ve.
- There are no x-Intercepts.
- Y-intercept $(0, 1)$.
- $y = 0$ is a horizontal asymp

one-one.

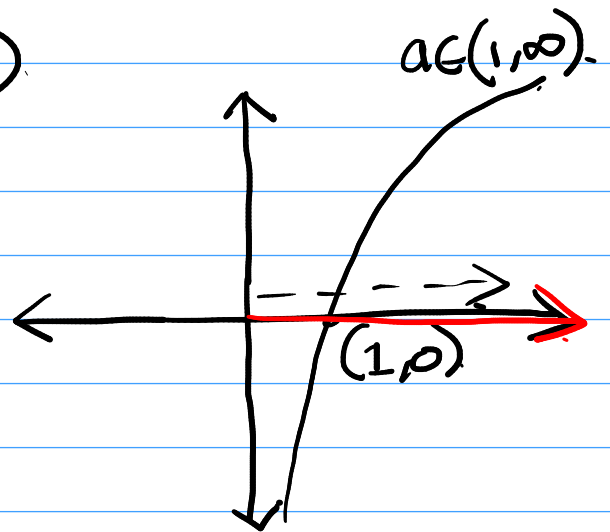
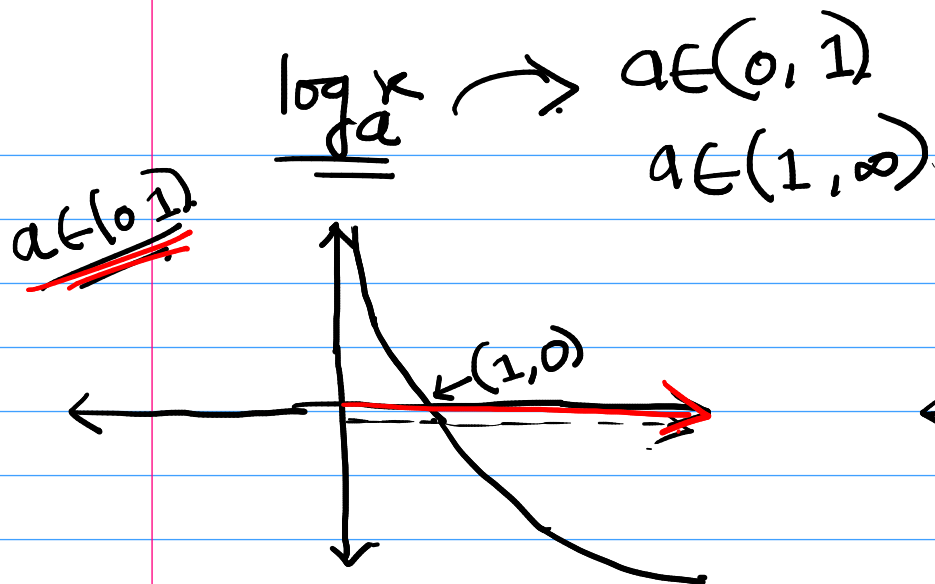
$f: \mathbb{R} \rightarrow \underline{(0, \infty)}$

$f(x) = a^x$ - one-one -

Inverse

$\log_a: \underline{(0, \infty)} \rightarrow \text{Range/Codomain}$
 $\leftarrow x \rightarrow \log_a x$

$\log_a x = y \Leftrightarrow \underline{a^y = x}$



• $f(x) = \log_{0.5}(x^2 - 3)$

$x^2 - 3 > 0 \neq 1 \in (0, \infty)$

$\Rightarrow x^2 > 3$

$x \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$ ← Domain(f)

• $g(x) = \ln\left(\frac{2x+3}{x^2-9}\right)$

$\frac{2x+3}{x^2-9} > 0$

$(-3, -3/2) \cup (3, \infty)$

$(3, \infty)$

$a > 0 \wedge b > 0$

$2x+3 > 0 \wedge x^2-9 > 0$

$\Rightarrow x > -3/2$

$\Rightarrow x \in (-3/2, \infty)$

$\frac{2x+3}{x^2-9} > 0$

$\frac{a}{b} > 0$

$a > 0 \wedge b > 0$

$a < 0 \wedge b < 0$

$2x+3 < 0 \wedge x^2-9 < 0$

$x < -3/2$

$x^2-9 < 0 \Rightarrow x \in (-3, 3)$

$x < -3/2 \wedge x \in (-3, 3) \Rightarrow x \in (-3, -3/2)$